

Answer Key

1. **b) BACD**
2. **b) 30**
3. **b) Some programmers are creative**
4. **b) 5:36 PM**
5. **d) HTML**
6. **a) 5 km** (*Pythagoras Theorem: $\sqrt{4^2 + 3^2} = 5$*)
7. **a) 24 and 12**
8. **b) 10 seconds** (*Speed = 72 km/h = 20 m/s, Time = 180 / 20 = 10 s*)
9. **c) 20 days** (*Work is inversely proportional to workers: $(6 \times 10) / 4 = 20$*)
10. **b) 126** (*$n^3 + 1$ pattern: $1^3+1, 2^3+1, 3^3+1...$*)
11. **b) ₹60 lakh** (*$50 + 20\% \text{ of } 50 = 60$*)
12. **b) 180** (*$30\% \text{ of } 600 = 180$*)
13. **b) 27.6** (*Average = $(20+25+30+28+35)/5 = 27.6$*)
14. **c) 60,000** (*50% increase of 40,000 = 60,000*)
15. **c) 15%** (*$100 - (65+25) = 10\% \text{ non-responders}$*)
16. **c) Bubble Sort** (*Worst-case time complexity $O(n^2)$*)
17. **b) 11001** (*Binary representation of 25 is 11001*)
18. **c) 3NF** (*3rd Normal Form removes transitive dependency*)
19. **c) PUT** (*PUT updates the entire resource, PATCH updates partially*)
20. **b) Stack** (*Follows LIFO principle: Last-In, First-Out*)
21. **b) DROP TABLE table_name;** (*Completely deletes table structure and data*)
22. **c) MongoDB** (*MongoDB is a NoSQL database*)
23. **b) Open Systems Interconnection** (*OSI Model = Open Systems Interconnection Model*)
24. **c) Router** (*Routers direct data packets between networks*)
25. **d) HTTPS** (*Hypertext Transfer Protocol Secure for encrypted web communication*)